

Day4_classAssignment1

Evaluation 1:

```
>_MONGOSH

> use day4
< switched to db day4
> db.createCollection("Books")
< { ok: 1 }
> db.createCollection("Purchase")
< { ok: 1 }
> db.Books.insertMany([
  {
    id:1,
    Title:"Data Structure",
    Author:"Lipschute",
    Subject:"DS",
    Publisher:"McGraw",
    Edition:"First Edition",
    Quantity:4,
    Price:217
  },
  {
    id:2,
    Title:"DOS Guide",
    Author:"NORTRON",
    Subject:"OS",
    Publisher:"PHI",
    Edition:"First Edition",
    Quantity:3,
    Price:175
  },
  {
    id:3,
```

>_MONGOSH

```
Title: "Turbo C++",
Author: "Robert Lafore",
Subject: "Prog",
Publisher: "Galgotia",
Edition: "Second Edition",
Quantity: 5,
Price: 270
},
{
  id: 4,
  Title: "Dbase Dummies",
  Author: "Palmer",
  Subject: "DBMS",
  Publisher: "PustakM",
  Edition: "First Edition",
  Quantity: 7,
  Price: 130
},
{
  id: 5,
  Title: "Mastering Windows",
  Author: "Cowart",
  Subject: "OS",
  Publisher: "BPB",
  Edition: "Third Edition",
  Quantity: 1,
  Price: 225
},
{
  id: 6,
```

>_MONGOSH

```
id:6,  
Title:"Computer Studies",  
Author:"French",  
Subject:"FND",  
Publisher:"Galgotia",  
Edition:"Second Edition",  
Quantity:2,  
Price:75  
,  
{  
id:7,  
Title:"COBOL",  
Author:"Stern",  
Subject:"Prog",  
Publisher:"John W",  
Edition:"Third Edition",  
Quantity:4,  
Price:1000  
,  
{  
id:8,  
Title:"Guide Network",  
Author:"Freed",  
Subject:"NET",  
Publisher:"Zpress",  
Edition:"Second Edition",  
Quantity:3,  
Price:200  
,
```

>_MONGOSH

```
{
  id:9,
  Title:"Basic for Beginners",
  Author:"Norton",
  Subject:"Prog",
  Publisher:"BPB",
  Edition:"First Edition",
  Quantity:3,
  Price:40
},
{
  id:10,
  Title:"Advanced Pascal",
  Author:"Schildt",
  Subject:"Prog",
  Publisher:"MCGraw",
  Edition:"Third Edition",
  Quantity:4,
  Price:350
}
])
```

```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a154294a17f95f10869660a'),
    '1': ObjectId('6a154294a17f95f10869660b'),
    '2': ObjectId('6a154294a17f95f10869660c'),
    '3': ObjectId('6a154294a17f95f10869660d'),
    '4': ObjectId('6a154294a17f95f10869660e'),
```

> MONGOSH

```
> db.Purchase.insertMany([
  {order_no:70001,Book_id:3,item_quantity:3,purchase_amt:810,date:"2022-11-05",customer_id:3005,salesman_id:5002},
  {order_no:70009,Book_id:4,item_quantity:4,purchase_amt:520,date:"2022-11-10",customer_id:3001,salesman_id:5005},
  {order_no:70002,Book_id:5,item_quantity:1,purchase_amt:225,date:"2022-11-09",customer_id:3002,salesman_id:5001},
  {order_no:70004,Book_id:6,item_quantity:2,purchase_amt:150,date:"2022-11-08",customer_id:3009,salesman_id:5003},
  {order_no:70007,Book_id:1,item_quantity:2,purchase_amt:434,date:"2022-11-08",customer_id:3005,salesman_id:5002},
  {order_no:70008,Book_id:8,item_quantity:1,purchase_amt:200,date:"2022-11-01",customer_id:3007,salesman_id:5001},
  {order_no:70010,Book_id:7,item_quantity:2,purchase_amt:2000,date:"2022-11-03",customer_id:3004,salesman_id:5006},
  {order_no:70003,Book_id:10,item_quantity:2,purchase_amt:700,date:"2022-11-04",customer_id:3009,salesman_id:5003},
  {order_no:70012,Book_id:9,item_quantity:2,purchase_amt:80,date:"2022-11-02",customer_id:3008,salesman_id:5002},
  {order_no:70011,Book_id:2,item_quantity:2,purchase_amt:350,date:"2022-11-07",customer_id:3003,salesman_id:5007},
  {order_no:70013,Book_id:7,item_quantity:2,purchase_amt:2000,date:"2022-11-06",customer_id:3002,salesman_id:5001}
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a15435ca17f95f108696614'),
    '1': ObjectId('6a15435ca17f95f108696615'),
    '2': ObjectId('6a15435ca17f95f108696616'),
    '3': ObjectId('6a15435ca17f95f108696617'),
    '4': ObjectId('6a15435ca17f95f108696618'),
    '5': ObjectId('6a15435ca17f95f108696619'),
    '6': ObjectId('6a15435ca17f95f10869661a'),
    '7': ObjectId('6a15435ca17f95f10869661b'),
    '8': ObjectId('6a15435ca17f95f10869661c'),
    '9': ObjectId('6a15435ca17f95f10869661d'),
    '10': ObjectId('6a15435ca17f95f10869661e')
  }
}
```

>_MONGOSH

```
> db.Books.createIndex({Title:"text"})
< Title_text
> db.Books.find(
  {$text:{$search:"COBOL"}}
)
< {
  _id: ObjectId('6a154294a17f95f108696610'),
  id: 7,
  Title: 'COBOL',
  Author: 'Stern',
  Subject: 'Prog',
  Publisher: 'John W',
  Edition: 'Third Edition',
  Quantity: 4,
  Price: 1000
}
```

>_MONGOSH

```
> db.Books.aggregate([
  {
    $group:{
      _id:"$Title",
      totalPrice:{$sum:"$Price"}
    }
  },
  {
    $sort:{_id:1}
  }
])
< {
  _id: 'Advanced Pascal',
  totalPrice: 350
}
{
  _id: 'Basic for Beginners',
  totalPrice: 40
}
{
  _id: 'COBOL',
  totalPrice: 1000
}
{
  _id: 'Computer Studies',
  totalPrice: 75
}
{
  _id: 'DOS Guide',
```

```
> db.Purchase.aggregate([
  {
    $group:{
      _id:"$Book_id",
      count:{$sum:1},
      totalQty:{$sum:"$item_quantity"}
    }
  },
  {
    $match:{count:{$gt:1}}
  }
])
< {
  _id: 7,
  count: 2,
  totalQty: 4
}
```



```
> db.Books.aggregate([
  {
    $group:{
      _id:"$Edition",
      totalPrice:{$sum:"$Price"}
    }
  }
])
< {
  _id: 'Third Edition',
  totalPrice: 1575
}
{
  _id: 'First Edition',
  totalPrice: 562
}
{
  _id: 'Second Edition',
  totalPrice: 545
}
```

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```
> db.Books.aggregate([
  {
    $project:{
      Title:1,
      totalAmount:{$multiply:["$Quantity","$Price"]}
    }
  },
  {
    $sort:{totalAmount:1}
  }
])
< {
  _id: ObjectId('6a154294a17f95f108696612'),
  Title: 'Basic for Beginners',
  totalAmount: 120
}
{
  _id: ObjectId('6a154294a17f95f10869660f'),
  Title: 'Computer Studies',
  totalAmount: 150
}
{
  _id: ObjectId('6a154294a17f95f10869660e'),
  Title: 'Mastering Windows',
  totalAmount: 225
}
{
  _id: ObjectId('6a154294a17f95f10869660b'),
  Title: 'DOS Guide',
```

```
> db.Purchase.aggregate([
  {
    $group:{
      _id:null,
      averagePurchase:{$avg:"$purchase_amt"}
    }
  }
])
< {
  _id: null,
  averagePurchase: 679
}
```

>_MONGOSH

```
> db.Books.aggregate([
  {
    $project:{
      Title:1,
      totalAmount:{$multiply:["$Quantity","$Price"]}
    },
    {
      $match:{
        totalAmount:{$gte:1000}
      }
    },
    {
      $sort:{totalAmount:-1}
    }
  ])
< {
  _id: ObjectId('6a154294a17f95f108696610'),
  Title: 'COBOL',
  totalAmount: 4000
}
{
  _id: ObjectId('6a154294a17f95f108696613'),
  Title: 'Advanced Pascal',
  totalAmount: 1400
}
{
  _id: ObjectId('6a154294a17f95f10869660c'),
  Title: 'Turbo C++',
```

```
>_MONGOSH
```

```
> let oldname="COBOL";  
   let newname="COBOL_new_Edition";  
  
   db.Books.updateOne(  
     {Title:oldname},  
     {$set:{Title:newname}}  
   )  
< {  
   acknowledged: true,  
   insertedId: null,  
   matchedCount: 1,  
   modifiedCount: 1,  
   upsertedCount: 0  
}
```

```
>_MONGOSH
```

```
> db.Books.createIndex(  
  {id:1},  
  {unique:true}  
)  
< id_1
```

>_MONGOSH

```
> db.createCollection("counters")
< { ok: 1 }
> db.counters.insertOne({
  _id:"bookid",
  sequence_value:10
})
< {
  acknowledged: true,
  insertedId: 'bookid'
}
> function getNextSequenceValue(sequenceName){
  var sequenceDocument = db.counters.findOneAndUpdate(
    {_id:sequenceName},
    {$inc:{sequence_value:1}},
    {returnNewDocument:true}
  );

  return sequenceDocument.sequence_value;
}
< [Function: getNextSequenceValue]
> db.Books.insertOne({
  id:getNextSequenceValue("bookid"),
  Title:"Java",
  Price:500
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a154706a17f95f10869661f')
}
```

>_MONGOSH

```
> db.Purchase.find({
  purchase_amt:{$mod:[4,0]}
})
< {
  _id: ObjectId('6a15435ca17f95f108696615'),
  order_no: 70009,
  Book_id: 4,
  item_quantity: 4,
  purchase_amt: 520,
  date: '2022-11-10',
  customer_id: 3001,
  salesman_id: 5005
}
{
  _id: ObjectId('6a15435ca17f95f108696619'),
  order_no: 70008,
  Book_id: 8,
  item_quantity: 1,
  purchase_amt: 200,
  date: '2022-11-01',
  customer_id: 3007,
  salesman_id: 5001
}
{
  _id: ObjectId('6a15435ca17f95f10869661a'),
  order_no: 70010,
  Book_id: 7,
  item_quantity: 2,
  purchase_amt: 2000,
```

Evaluation 2:

>_MONGOSH

```
> db.createCollection("Customer")
```

```
< { ok: 1 }
```

```
> db.Customer.insertMany([
```

```
{
```

```
  id:3001,
```

```
  FirstName:"Ben",
```

```
  LastName:"Miller",
```

```
  Street:"101,Candy Rd",
```

```
  City:"Redmond",
```

```
  State:"WA",
```

```
  Phone:9845367123,
```

```
  Email:"ben@gmail.com"
```

```
},
```

```
{
```

```
  id:3002,
```

```
  FirstName:"Garrett",
```

```
  LastName:"Vargas",
```

```
  Street:"102, Acorn Avenue",
```

```
  City:"Calgary",
```

```
  State:"AB",
```

```
  Phone:9789011111,
```

```
  Email:"garrett@gmail.com"
```

```
},
```

```
{
```

```
  id:3003,
```

```
  FirstName:"Gabe",
```

```
  LastName:"Mares",
```

```
  Street:"1061, Buskrit Avenue",
```

```
  City:"Edmonds",
```


>_MONGOSH

```
{
  id:3004,
  FirstName:"Reuben",
  LastName:"Dsa",
  Street:"1064, Slow Creek Road",
  City:"Seattle",
  State:"WA",
  Phone:9988844555,
  Email:"reuben@gmail.com"
},
{
  id:3005,
  FirstName:"Gordon",
  LastName:"Hee",
  Street:"108, LakeSide Court",
  City:"Bellevue",
  State:"WA",
  Phone:8855577234,
  Email:"gordon@gmail.com"
},
{
  id:3006,
  FirstName:"Karan",
  LastName:"Khanna",
  Street:"1102, Ravenwood",
  City:"Seattle",
  State:"WA",
  Phone:7788966445,
  Email:"karan@gmail.com"
},
```

>_MONGOSH

```
{
  id:3007,
  FirstName:"Francois",
  LastName:"Ajenstat",
  Street:"1144, Paradise Ct",
  City:"Issaquah",
  State:"WA",
  Phone:7889900123,
  Email:"francois@gmail.com"
},
{
  id:3008,
  FirstName:"Sariya",
  LastName:"Hampa",
  Street:"1185, Dallas Drive",
  City:"Everett",
  State:"WA",
  Phone:9667785544,
  Email:"sariya@gmail.com"
},
{
  id:3009,
  FirstName:"Krik",
  LastName:"Koenigs",
  Street:"1220, Bradford Way",
  City:"Seattle",
  State:"WA",
  Phone:9774455443,
  Email:"krik@gmail.com"
},
```

>_MONGOSH

```
Phone:9774455443,
Email:"krik@gmail.com"
},
{
  id:3010,
  FirstName:"Kim",
  LastName:"Ralls",
  Street:"1226, Shoe St",
  City:"Bothell",
  State:"WA",
  Phone:8770033221,
  Email:"kim@gmail.com"
}
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a1549eca17f95f108696620'),
    '1': ObjectId('6a1549eca17f95f108696621'),
    '2': ObjectId('6a1549eca17f95f108696622'),
    '3': ObjectId('6a1549eca17f95f108696623'),
    '4': ObjectId('6a1549eca17f95f108696624'),
    '5': ObjectId('6a1549eca17f95f108696625'),
    '6': ObjectId('6a1549eca17f95f108696626'),
    '7': ObjectId('6a1549eca17f95f108696627'),
    '8': ObjectId('6a1549eca17f95f108696628'),
    '9': ObjectId('6a1549eca17f95f108696629')
  }
}
```

```
>_MONGOSH
```

```
> db.createCollection("Salesman")
```

```
< { ok: 1 }
```

```
> db.Salesman.insertMany([
```

```
{
```

```
  id:5001,
```

```
  Name:"James Hoog",
```

```
  City:"New York",
```

```
  Commission:0.15
```

```
},
```

```
{
```

```
  id:5002,
```

```
  Name:"Nail Knite",
```

```
  City:"Paris",
```

```
  Commission:0.13
```

```
},
```

```
{
```

```
  id:5003,
```

```
  Name:"Lauson Hen",
```

```
  City:"Germany",
```

```
  Commission:0.12
```

```
},
```

```
{
```

```
  id:5005,
```

```
  Name:"Pit Alex",
```

```
  City:"London",
```

```
  Commission:0.11
```

```
},
```

```
{
```

```
  id:5006,
```

>_MONGOSH

```
{
  id:5005,
  Name:"Pit Alex",
  City:"London",
  Commission:0.11
},
{
  id:5006,
  Name:"Mc Lyon",
  City:"Paris",
  Commission:0.14
},
{
  id:5007,
  Name:"Paul Adam",
  City:"Rome",
  Commission:0.13
}
])
```

```
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a154ab4a17f95f10869662a'),
    '1': ObjectId('6a154ab4a17f95f10869662b'),
    '2': ObjectId('6a154ab4a17f95f10869662c'),
    '3': ObjectId('6a154ab4a17f95f10869662d'),
    '4': ObjectId('6a154ab4a17f95f10869662e'),
    '5': ObjectId('6a154ab4a17f95f10869662f')
  }
}
```

>_MONGOSH

```
> db.Purchase.aggregate([
  {
    $lookup:{
      from:"Salesman",
      localField:"salesman_id",
      foreignField:"id",
      as:"salesman_details"
    }
  }
])
< {
  _id: ObjectId('6a15435ca17f95f108696614'),
  order_no: 70001,
  Book_id: 3,
  item_quantity: 3,
  purchase_amt: 810,
  date: '2022-11-05',
  customer_id: 3005,
  salesman_id: 5002,
  salesman_details: [
    {
      _id: ObjectId('6a154ab4a17f95f10869662b'),
      id: 5002,
      Name: 'Nail Knite',
      City: 'Paris',
      Commission: 0.13
    }
  ]
}
```

>_MONGOSH

```
> db.Customer.aggregate([
  {
    $match:{
      State:"WA"
    }
  },
  {
    $project:{
      _id:0,
      FirstName:1,
      LastName:1,
      City:1,
      State:1
    }
  },
  {
    $sort:{
      City:1
    }
  }
])
< {
  FirstName: 'Gordon',
  LastName: 'Hee',
  City: 'Bellevue',
  State: 'WA'
}
{
  FirstName: 'Kim',
```

>_MONGOSH

```
> db.Salesman.updateOne(
  {id:5005},
  {
    $set:{
      Commission:0.13
    }
  }
)
```

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
> db.Salesman.aggregate([
  {
    $match:{
      id:5005
    }
  }
])
```

```
< {
  _id: ObjectId('6a154ab4a17f95f10869662d'),
  id: 5005,
  Name: 'Pit Alex',
  City: 'London',
  Commission: 0.13
}
```


>_MONGOSH

```
> db.Customer.aggregate([
  {
    $match:{
      FirstName:/^G/
    }
  }
])
< {
  _id: ObjectId('6a1549eca17f95f108696621'),
  id: 3002,
  FirstName: 'Garrett',
  LastName: 'Vargas',
  Street: '102, Acorn Avenue',
  City: 'Calgary',
  State: 'AB',
  Phone: 9789011111,
  Email: 'garrett@gmail.com'
}
{
  _id: ObjectId('6a1549eca17f95f108696622'),
  id: 3003,
  FirstName: 'Gabe',
  LastName: 'Mares',
  Street: '1061, Buskrit Avenue',
  City: 'Edmonds',
  State: 'WA',
  Phone: 9944433222,
  Email: 'gabe@gmail.com'
}
```

>_MONGOSH

```
> db.Salesman.createIndex({  
  Name:"text"  
})
```

```
< Name_text
```

```
> db.Salesman.aggregate([  
  {  
    $match:{  
      $text:{  
        $search:"Pit Alex"  
      }  
    }  
  }  
])
```

```
< {  
  _id: ObjectId('6a154ab4a17f95f10869662d'),  
  id: 5005,  
  Name: 'Pit Alex',  
  City: 'London',  
  Commission: 0.13  
}
```

>_MONGOSH

```
> db.Customer.aggregate([
  {
    $match:{
      FirstName:{
        $regex:"^k",
        $options:"i"
      }
    }
  }
])
< {
  _id: ObjectId('6a1549eca17f95f108696625'),
  id: 3006,
  FirstName: 'Karan',
  LastName: 'Khanna',
  Street: '1102, Ravenwood',
  City: 'Seattle',
  State: 'WA',
  Phone: 7788966445,
  Email: 'karan@gmail.com'
}
{
  _id: ObjectId('6a1549eca17f95f108696628'),
  id: 3009,
  FirstName: 'Krik',
  LastName: 'Koenigs',
  Street: '1220, Bradford Way',
  City: 'Seattle',
  State: 'WA',
```

>_MONGOSH

```
> db.Customer.aggregate([
  {
    $sort:{
      _id:-1
    }
  },
  {
    $skip:2
  },
  {
    $limit:1
  }
])
< {
  _id: ObjectId('6a1549eca17f95f108696627'),
  id: 3008,
  FirstName: 'Sariya',
  LastName: 'Hampa',
  Street: '1185, Dallas Drive',
  City: 'Everett',
  State: 'WA',
  Phone: 9667785544,
  Email: 'sariya@gmail.com'
}
```

>_MONGOSH

```
> db.Purchase.aggregate([
  {
    $lookup:{
      from:"Salesman",
      localField:"salesman_id",
      foreignField:"id",
      as:"salesman"
    }
  }
])
< {
  _id: ObjectId('6a15435ca17f95f108696614'),
  order_no: 70001,
  Book_id: 3,
  item_quantity: 3,
  purchase_amt: 810,
  date: '2022-11-05',
  customer_id: 3005,
  salesman_id: 5002,
  salesman: [
    {
      _id: ObjectId('6a154ab4a17f95f10869662b'),
      id: 5002,
      Name: 'Nail Knite',
      City: 'Paris',
      Commission: 0.13
    }
  ]
}
```

>_MONGOSH

```
> db.Customer.aggregate([
  {
    $lookup:{
      from:"Purchase",
      localField:"id",
      foreignField:"customer_id",
      as:"purchase_details"
    }
  },
  {
    $unwind:"$purchase_details"
  },
  {
    $project:{
      _id:0,
      FirstName:1,
      LastName:1,
      Phone:1,
      Email:1,
      item_quantity:"$purchase_details.item_quantity",
      purchase_amt:"$purchase_details.purchase_amt"
    }
  },
  {
    $sort:{FirstName:1}
  }
])
< {
  FirstName: 'Ben',
  LastName: 'Miller',
  Phone: 9845367123,
  Email: 'ben@gmail.com',
  item_quantity: 4,
  purchase_amt: 520
}
```

```
>_MONGOSH
```

```
>
```

```
> db.createCollection("students3")
```

```
< { ok: 1 }
```

```
> db.students3.insertMany([
```

```
{
```

```
  "_id":1,
```

```
  "tests":[95,92,90],
```

```
  "average":92,
```

```
  "grade":"A",
```

```
  "lastUpdate":ISODate("2020-01-23T05:18:40.013Z")
```

```
},
```

```
{
```

```
  "_id":2,
```

```
  "tests":[70,75,82],
```

```
  "lastUpdate":ISODate("2019-01-01T00:00:00Z")
```

```
}
```

```
])
```

```
< {
```

```
  acknowledged: true,
```

```
  insertedIds: {
```

```
    '0': 1,
```

```
    '1': 2
```

```
  }
```

```
}
```

>_MONGOSH

```
> db.students3.updateOne(
  { _id: 2 },
  [
    {
      $set: {
        average: { $avg: "$tests" }
      }
    },
    {
      $set: {
        grade: {
          $switch: {
            branches: [
              {
                case: { $gte: [ "$average", 90 ] },
                then: "A"
              },
              {
                case: { $gte: [ "$average", 80 ] },
                then: "B"
              },
              {
                case: { $gte: [ "$average", 70 ] },
                then: "C"
              }
            ],
            default: "D"
          }
        }
      },
      $set: {
        lastUpdate: new Date()
      }
    }
  ]
)
< {
```

ok: 1 updated: 1

>_MONGOSH

> db.students3.find({_id:2})

```
< {
  _id: 2,
  tests: [
    70,
    75,
    82
  ],
  lastUpdate: 2026-05-26T07:59:00.786Z,
  average: 75.66666666666667,
  grade: 'C'
}
```